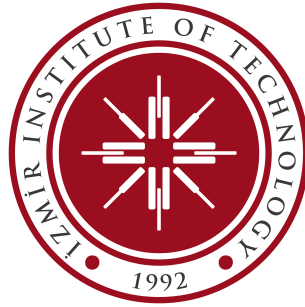




İzmir Institute of Technology
Faculty of Engineering
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EE 315 Electronics Laboratory Prelab Sheet

Experiment 1

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Problem 1

1.a) *When measuring a signal, under what circumstances should each type of voltage value, V_p , V_{pp} , V_{rms} , be used?*

- **Peak Voltage (V_p):** Used for determining the maximum voltage levels relative to ground. Example use cases include determining diode's reverse voltage rating or finding the maximum output of a rectifier or clipper. For asymmetric waveforms relative to ground, the positive and negative peaks must be measured separately.
- **Peak-to-Peak Voltage (V_{pp}):** Used when the total swing of the signal is required. It does not depend on the DC offset or the ground reference. For instance, function generators are typically set in terms of V_{pp} . Additionally, it is the standard metric for measuring ripple or noise amplitude.
- **RMS Voltage (V_{rms}):** Used for power calculation and thermal analysis as it represents the equivalent DC voltage which delivers the same average power to load. It is also the best metric for comparing waveforms with different shapes.

1.b) *How does the purpose of measurement (e.g., power analysis, amplitude comparison, waveform characterization) influence this choice?*

- **Power analysis:** For power analysis, RMS is the right choice. If the total power is needed, the DC part must be included (AC+DC RMS). If only the ripple is of interest, the AC RMS is enough.
- **Amplitude comparison:** For amplitude comparison, V_{pp} or V_p is easier because it can be read directly from the oscilloscope screen, but this only works well if the waveforms have the same shape.
- **Waveform characterization:** For waveform characterization, one value is not enough. We would measure the peaks, V_{pp} , the DC value, the RMS value, and the timing (period, duty cycle, turn-on/turn-off times). The crest factor and ripple factor can then be calculated from these.

1.c) *What are the differences between how oscilloscopes, multimeters, and simulation tools measure RMS or peak values?*

The three tools work differently:

- **Oscilloscope:** It samples the signal with an ADC. It finds V_{max} , V_{min} and V_{pp} from the samples and calculates RMS numerically as $\sqrt{\sum v^2/N}$. Because it uses the actual samples, its RMS is correct for any waveform shape. The "cycle RMS" option, which uses whole periods, is the most accurate.
- **Multimeter:** Cheaper meters are average-responding. They rectify the signal, take the average, and multiply by 1.11, so they are only correct for sine waves. True-RMS meters calculate the real RMS but have limited bandwidth and crest factor. In AC mode they block the DC part. Multimeters usually cannot measure peak values.
- **Simulation (LTspice):** The values are calculated directly from their definitions using .meas or cursors. There is no noise and no bandwidth limit. However, the results depend on the time step and on the diode model, and RMS has to be taken over full periods after the transients have settled.

1.d) Which limitations (sampling rate, bandwidth, waveform type, numerical filtering, etc.) can affect the accuracy or validity of your selected measurement method?

Several limitations can affect accuracy:

- **Sampling rate:** If it is too low, short peaks can be missed or the signal can alias.
- **Bandwidth:** The higher harmonics of square and pulse waves are attenuated, so the edges and peaks become rounded.
- **Waveform type:** Average-responding meters give wrong results for non-sinusoidal waves, and signals with a high crest factor can exceed the meter's limits.
- **Coupling:** AC coupling removes the DC component and can distort low-frequency signals.
- **Measurement window:** RMS calculated over a non-integer number of periods gives an error.
- **Noise:** Spikes affect the peak and V_{pp} readings more than the RMS value.
- **Scope resolution:** The scope has only 8 bits, so the signal should fill the screen.
- **Probes:** The probe setting ($\times 1$ or $\times 10$) must be correct. Voltages across components that are not grounded need the two-probe differential method, which adds some error.
- **Simulation:** The time step size and the accuracy of the model affect the results.

1.e) Based on your discussion in 1.a), 1.b), 1.c), and 1.d), justify which measurement technique is the most appropriate for measuring V_p , V_{pp} and V_{rms} . Explain your reasoning by referring to the characteristics of each quantity, the type of waveform, and the capabilities of your measurement instruments.

- **For V_p and V_{pp} :** The oscilloscope with DC coupling is the most appropriate method. These are instantaneous values, so we need to see the waveform, and a multimeter cannot measure them. Our signals are around 100Hz, so bandwidth and sampling rate are not a problem. DC coupling keeps the real peak values of the asymmetric clipper and rectifier outputs. We would check the automatic measurements with cursors, and use CH1 – CH2 for components that are not grounded.
- **For V_{rms} :** We would use the oscilloscope's cycle RMS measurement, or a true-RMS multimeter, which is fine at 100Hz. The waveforms in this lab (triangular, clipped, rectified) are not sine waves, so an average-responding meter would give wrong values. If the multimeter is in AC mode, we would also measure the DC value and calculate the total RMS as $\sqrt{V_{DC}^2 + V_{ACrms}^2}$.

Problem 2

2.a) Show that the RMS value V_{rms} of a sinusoidal voltage waveform $v(t) = V_p \sin(\omega t)$ is given by

$$V_{\text{rms}} = \frac{V_p}{\sqrt{2}},$$

Where V_p is the peak value of the voltage waveform.

The root-mean-square (RMS) value of a continuous periodic voltage waveform $v(t)$ of period T is defined by:

$$V_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$$

For a sinusoidal voltage $v(t) = V_p \sin(\omega t)$, the function $v^2(t) = V_p^2 \sin^2(\omega t)$ is periodic with period $T/2$. Due to half-cycle symmetry, the mean-square value over the full period equals the average over a half period:

$$V_{\text{rms}} = \sqrt{\frac{2}{T} \int_0^{T/2} V_p^2 \sin^2(\omega t) dt}$$

Using the power-reduction trigonometric identity:

$$\sin^2(\omega t) = \frac{1 - \cos(2\omega t)}{2}$$

Substituting this identity into the integral:

$$\frac{2}{T} \int_0^{T/2} V_p^2 \sin^2(\omega t) dt = \frac{2V_p^2}{T} \int_0^{T/2} \frac{1 - \cos(2\omega t)}{2} dt \quad (1)$$

$$= \frac{V_p^2}{T} \int_0^{T/2} (1 - \cos(2\omega t)) dt \quad (2)$$

Integrating term by term:

$$\int_0^{T/2} (1 - \cos(2\omega t)) dt = \left[t - \frac{\sin(2\omega t)}{2\omega} \right]_0^{T/2}$$

Recalling that the angular frequency is $\omega = \frac{2\pi}{T}$, the term inside the sine function evaluated at $t = T/2$ becomes:

$$2\omega \left(\frac{T}{2} \right) = \omega T = \left(\frac{2\pi}{T} \right) T = 2\pi$$

Evaluating between the boundaries:

$$\left[t - \frac{\sin(2\omega t)}{2\omega} \right]_0^{T/2} = \left(\frac{T}{2} - \frac{\sin(2\pi)}{2\omega} \right) - \left(0 - \frac{\sin(0)}{2\omega} \right) \quad (3)$$

$$= \frac{T}{2} - 0 = \frac{T}{2} \quad (4)$$

Substituting this evaluation back into the mean-square equation gives:

$$\frac{V_p^2}{T} \cdot \frac{T}{2} = \frac{V_p^2}{2}$$

Finally, taking the square root gives the RMS voltage:

$$V_{\text{rms}} = \sqrt{\frac{V_p^2}{2}} = \frac{V_p}{\sqrt{2}}$$

Problem 3

3.a) Show that the RMS value $V_{\Sigma\text{rms}}$ of a voltage waveform $v(t) = V_{\text{DC}} + v_{\text{AC}}(t)$ is given by

$$V_{\Sigma\text{rms}} = \sqrt{V_{\text{DC}}^2 + V_{\text{ACrms}}^2}$$

Where V_{ACrms} is the RMS value of the pure AC component $v_{\text{AC}}(t)$

The total root-mean-square (RMS) value $V_{\Sigma\text{rms}}$ of a periodic waveform $v(t)$ over a fundamental period T is defined by:

$$V_{\Sigma\text{rms}} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$$

For a waveform composed of a constant DC offset V_{DC} and a time-varying AC component $v_{\text{AC}}(t)$:

$$v(t) = V_{\text{DC}} + v_{\text{AC}}(t)$$

Squaring the voltage function yields:

$$v^2(t) = (V_{\text{DC}} + v_{\text{AC}}(t))^2 = V_{\text{DC}}^2 + 2V_{\text{DC}}v_{\text{AC}}(t) + v_{\text{AC}}^2(t)$$

Substituting this expansion into the mean-square expression:

$$\begin{aligned} \frac{1}{T} \int_0^T v^2(t) dt &= \frac{1}{T} \int_0^T (V_{\text{DC}}^2 + 2V_{\text{DC}}v_{\text{AC}}(t) + v_{\text{AC}}^2(t)) dt \\ &= \frac{1}{T} \int_0^T V_{\text{DC}}^2 dt + \frac{2V_{\text{DC}}}{T} \int_0^T v_{\text{AC}}(t) dt + \frac{1}{T} \int_0^T v_{\text{AC}}^2(t) dt \end{aligned}$$

Evaluating each integral term individually:

1. The DC constant term integrates directly:

$$\frac{1}{T} \int_0^T V_{\text{DC}}^2 dt = \frac{V_{\text{DC}}^2}{T} [t]_0^T = V_{\text{DC}}^2$$

2. By definition, the pure AC component $v_{\text{AC}}(t)$ has a zero time-average (DC) value over its period:

$$\frac{1}{T} \int_0^T v_{\text{AC}}(t) dt = 0$$

Therefore, the cross-product term vanishes:

$$\frac{2V_{\text{DC}}}{T} \int_0^T v_{\text{AC}}(t) dt = 0$$

3. The third term represents the square of the RMS value of the AC component by definition:

$$\frac{1}{T} \int_0^T v_{\text{AC}}^2(t) dt = V_{\text{ACrms}}^2$$

Combining the three evaluated components:

$$\frac{1}{T} \int_0^T v^2(t) dt = V_{\text{DC}}^2 + 0 + V_{\text{ACrms}}^2 = V_{\text{DC}}^2 + V_{\text{ACrms}}^2$$

Taking the square root of both sides gives the total RMS value:

$$V_{\Sigma\text{rms}} = \sqrt{V_{\text{DC}}^2 + V_{\text{ACrms}}^2}$$

Problem 4

4.a) Draw the output waveform for the circuit given below using the basic diode model with $V_{ON} = 0.6$ V. The input voltage v_{in} is a 100 Hz, 10 V_{p-p} triangular waveform. Note that the polarity of D_1 is reversed in this circuit. Calculate the peak output voltages at v_o and the turn-ON and turn-OFF times of the two diodes during one period of the input.

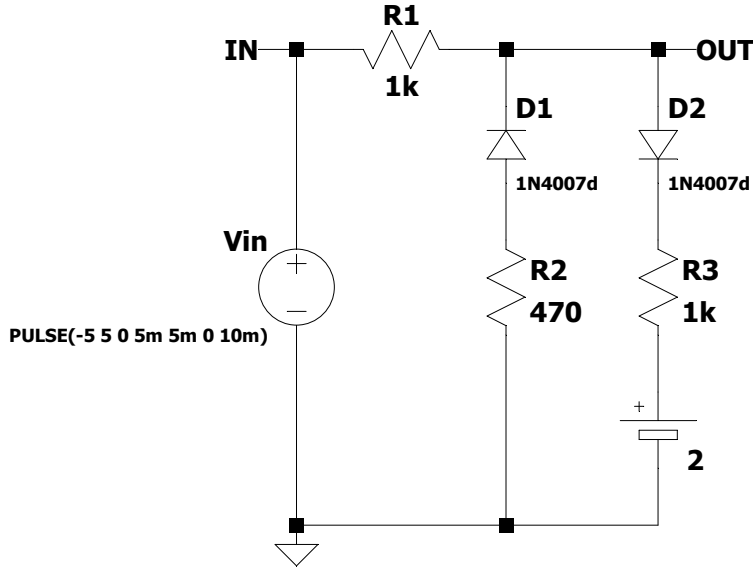


Figure 1: Circuit schematic for Problem 4.

1. Mathematical Description of the Input Waveform

The input signal $v_{in}(t)$ is a 100 Hz, 10 V_{p-p} triangular waveform symmetric around zero ground with a fundamental period of:

$$T = \frac{1}{f} = \frac{1}{100 \text{ Hz}} = 10 \text{ ms} \quad (1)$$

Defining one period starting at the negative peak ($t = 0$, $v_{in} = -5$ V):

$$v_{in}(t) = \begin{cases} -5 + 2t & \text{for } 0 \leq t \leq 5 \text{ ms} \quad (\text{Rising slope: } +2 \text{ V/ms}) \\ 15 - 2t & \text{for } 5 \text{ ms} \leq t \leq 10 \text{ ms} \quad (\text{Falling slope: } -2 \text{ V/ms}) \end{cases} \quad (2)$$

2. Piecewise Circuit Analysis

Using the constant-voltage diode model ($V_{ON} = 0.6$ V):

Region I: Both Diodes OFF ($-0.6 \text{ V} \leq v_{in} \leq +2.6 \text{ V}$) Diode D_1 is reverse-biased when $v_o > -0.6$ V, and D_2 is reverse-biased when $v_o < 2 \text{ V} + 0.6 \text{ V} = +2.6$ V. Because both diode branches are open-circuited, no current flows through R_1 :

$$v_o(t) = v_{in}(t) \quad (3)$$

Region II: Diode D_2 ON, D_1 OFF ($v_{in} > +2.6$ V) When $v_{in} > +2.6$ V, diode D_2 turns ON while D_1 remains OFF. Applying Kirchhoff's Current Law at node v_o :

$$\frac{v_{in} - v_o}{R_1} = \frac{v_o - (V_{ON} + 2 \text{ V})}{R_3} \quad (4)$$

With $R_1 = 1 \text{ k}\Omega$ and $R_3 = 1 \text{ k}\Omega$:

$$v_o(t) = \frac{v_{in}(t) + 2.6}{2} = 0.5v_{in}(t) + 1.3 \text{ V} \quad (5)$$

Region III: Diode D_1 ON, D_2 OFF ($v_{in} < -0.6$ V) When $v_{in} < -0.6$ V, diode D_1 turns ON with its cathode at v_o , while D_2 is OFF. Writing KCL at node v_o with current flowing from ground through R_2 and D_1 :

$$\frac{0 - V_{ON} - v_o}{R_2} = \frac{v_o - v_{in}}{R_1} \quad (6)$$

Substituting $R_1 = 1000\Omega$, $R_2 = 470\Omega$, and $V_{ON} = 0.6$ V:

$$1000(-0.6 - v_o) = 470(v_o - v_{in}) \implies 1470v_o = 470v_{in} - 600 \quad (7)$$

$$v_o(t) = \frac{47}{147}v_{in}(t) - \frac{20}{49} \approx 0.3197v_{in}(t) - 0.4082 \text{ V} \quad (8)$$

3. Peak Output Voltages

- **Positive Peak ($v_{o,max}$):** Occurs at $v_{in} = +5$ V:

$$v_{o,max} = 0.5(5 \text{ V}) + 1.3 \text{ V} = +\mathbf{3.80 \text{ V}} \quad (9)$$

- **Negative Peak ($v_{o,min}$):** Occurs at $v_{in} = -5$ V:

$$v_{o,min} = \frac{47}{147}(-5 \text{ V}) - \frac{20}{49} = -\frac{295}{147} \text{ V} \approx -\mathbf{2.01 \text{ V}} \quad (10)$$

4. Diode Switching Times (Turn-ON and Turn-OFF)

Within the period $t \in [0, 10 \text{ ms}]$:

1. **D_1 Turn-OFF time (t_1):** On the rising slope when $v_{in}(t_1) = -0.6$ V:

$$-5 + 2t_1 = -0.6 \implies 2t_1 = 4.4 \implies t_1 = \mathbf{2.20 \text{ ms}} \quad (11)$$

2. **D_2 Turn-ON time (t_2):** On the rising slope when $v_{in}(t_2) = +2.6$ V:

$$-5 + 2t_2 = 2.6 \implies 2t_2 = 7.6 \implies t_2 = \mathbf{3.80 \text{ ms}} \quad (12)$$

3. **D_2 Turn-OFF time (t_3):** On the falling slope when $v_{in}(t_3) = +2.6$ V:

$$15 - 2t_3 = 2.6 \implies 2t_3 = 12.4 \implies t_3 = \mathbf{6.20 \text{ ms}} \quad (13)$$

4. **D_1 Turn-ON time (t_4):** On the falling slope when $v_{in}(t_4) = -0.6$ V:

$$15 - 2t_4 = -0.6 \implies 2t_4 = 15.6 \implies t_4 = \mathbf{7.80 \text{ ms}} \quad (14)$$

In summary:

- Diode D_1 conducts for $0 \leq t \leq 2.20 \text{ ms}$ and $7.80 \text{ ms} \leq t \leq 10.00 \text{ ms}$ (Total ON time = 4.40 ms).
- Diode D_2 conducts for $3.80 \text{ ms} \leq t \leq 6.20 \text{ ms}$ (Total ON time = 2.40 ms).

4.b) Simulate the circuit in LTspice and verify your analytical results. Use 1N4007 diode models for D_1 and D_2 . A triangular input can be generated using the PULSE function. Set the source to vary between -5 V and $+5\text{ V}$ with $T_{\text{rise}} = 5\text{ ms}$, $T_{\text{fall}} = 5\text{ ms}$, $T_{\text{on}} = 0$, and $T_{\text{period}} = 10\text{ ms}$ to obtain a 100 Hz , $10\text{ V}_{\text{p-p}}$ triangular waveform. Run a transient simulation and:

- (i) Plot v_{in} and v_o on the same graph. Verify the positive and negative peak values of v_o obtained in Question 4.a:

The circuit shown in Figure 1 was simulated in LTspice using a transient simulation command (.tran 10m). The triangular waveform source was specified using parameters PULSE(-5V 5V 0s 5ms 5ms 0s 10ms) with 1N4007 diode models specified for D_1 and D_2 .

Figure 2 shows the simulated input voltage $v_{\text{in}}(t)$ and output voltage $v_o(t)$ plotted together over one full period ($0 \leq t \leq 10\text{ ms}$).

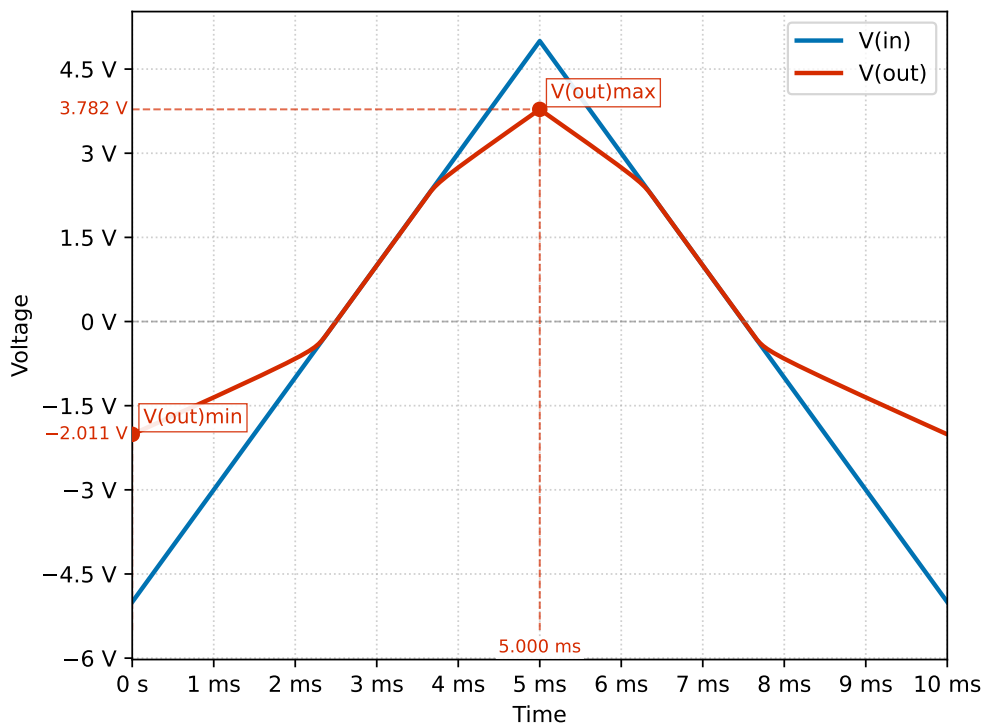


Figure 2: Simulated input voltage v_{in} and output voltage v_o waveforms.

Using the waveform cursors in LTspice (marked in Figure 2):

- **Positive Peak Verification ($v_{o,\text{max}}$):** At $t = 5.000\text{ ms}$ ($v_{\text{in}} = +5.00\text{ V}$), the simulated cursor measurement is $v_{o,\text{max},\text{sim}} = +3.782\text{ V}$. Comparing with the analytical value $v_{o,\text{max},\text{analytic}} = +3.80\text{ V}$:

$$\text{Relative Error} = \frac{|3.782\text{ V} - 3.80\text{ V}|}{3.80\text{ V}} \times 100\% = 0.47\%$$

- **Negative Peak Verification ($v_{o,\text{min}}$):** At $t = 0\text{ s}$ and $t = 10.000\text{ ms}$ ($v_{\text{in}} = -5.00\text{ V}$), the simulated cursor measurement is $v_{o,\text{min},\text{sim}} = -2.011\text{ V}$. Comparing with the analytical value $v_{o,\text{min},\text{analytic}} \approx -2.007\text{ V}$ ($-\frac{295}{147}\text{ V}$):

$$\text{Relative Error} = \frac{|-2.011\text{ V} - (-2.007\text{ V})|}{2.007\text{ V}} \times 100\% = 0.20\%$$

Both simulated peak voltages verify the analytical derivations from Question 4.a with less than 0.5% error.

- (ii) **Plot the diode currents i_{D1} and i_{D2} . Determine the turn-ON and turn-OFF times and compare with analytical results:**

The simulated branch currents through diodes D_1 and D_2 are plotted in Figure 3.

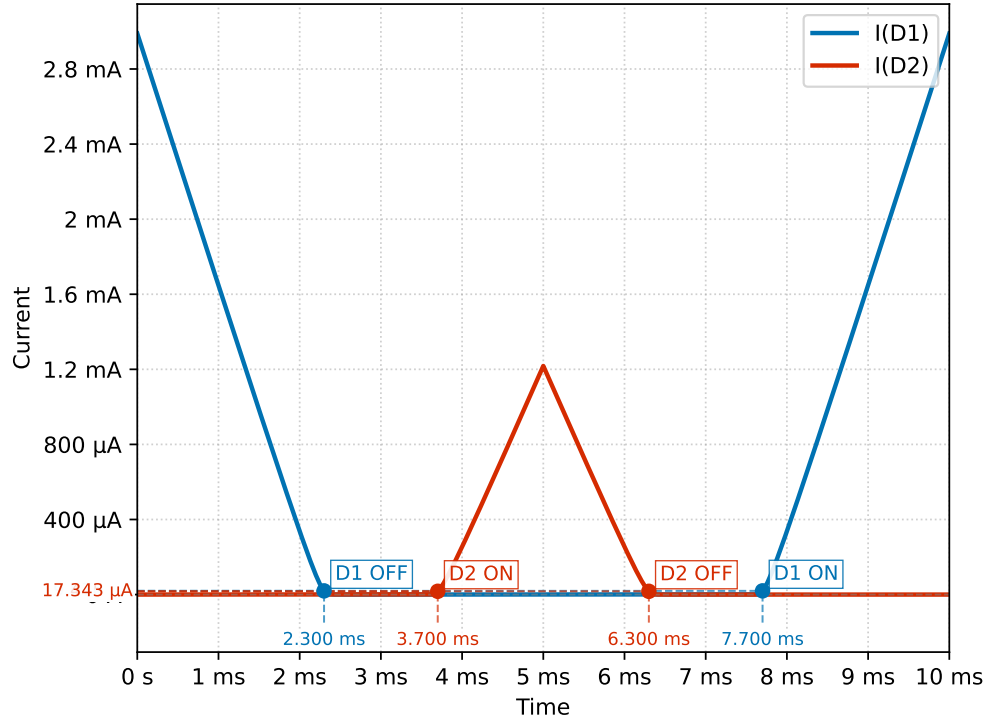


Figure 3: Simulated diode currents i_{D1} and i_{D2} with switching time cursors.